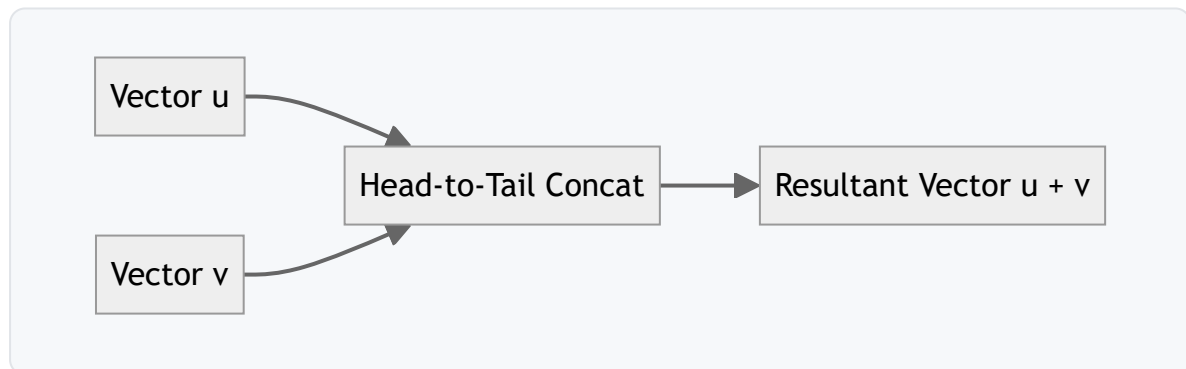


Day 21 - Lecture 21.8: Vector Addition & Geometric Laws

[Python 3.10+](#)[Jupyter Notebook](#)[PDF notes.pdf](#)[Language: English | हिंदी / Hinglish | !\[\]\(e3f8612927870f2e0f9f5989e6dd3064_img.jpg\) Back to Lecture 21 Module](#)

1. Overview & Component-wise Synthesis

Vector addition is the fundamental linear combination operation in vector spaces. Geometrically, it represents sequential spatial displacement. In Machine Learning, vector addition executes residual connections ($x + f(x)$ in ResNets), word embedding semantics ($\text{vec}\{\text{king}\} - \text{vec}\{\text{man}\} + \text{vec}\{\text{woman}\} \approx \text{vec}\{\text{queen}\}$), and gradient parameter updates.



2. Mathematical Formalism

1. Component-wise Algebraic Definition:

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 & u_2 & \vdots & u_n \end{bmatrix} + \begin{bmatrix} v_1 & v_2 & \vdots & v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 & u_2 + v_2 & \vdots & u_n + v_n \end{bmatrix}$$

2. Geometric Laws of Addition:

1. **Triangle Law of Addition (Head-to-Tail):** Place the tail of $\mathbf{v}$ at the head of $\mathbf{u}$. The vector from the tail of $\mathbf{u}$ to the head of $\mathbf{v}$ is $\mathbf{u} + \mathbf{v}$.
2. **Parallelogram Law of Addition:** If $\mathbf{u}$ and $\mathbf{v}$ emanate from the same origin, the resultant $\mathbf{u} + \mathbf{v}$ is the diagonal of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$.

3. Triangle Inequality Theorem:

The length of any side of a triangle cannot exceed the sum of the lengths of the other two sides:

$$|\mathbf{u} + \mathbf{v}| \leq |\mathbf{u}| + |\mathbf{v}|$$

Equality $|\mathbf{u} + \mathbf{v}| = |\mathbf{u}| + |\mathbf{v}|$ holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are collinear and point in the identical direction.

3. Python Implementation

```
import numpy as np

u = np.array([3.0, 1.0])
v = np.array([1.0, 4.0])

# Resultant vector
w = u + v

norm_u = np.linalg.norm(u)
norm_v = np.linalg.norm(v)
norm_w = np.linalg.norm(w)

print(f"Vector u: {u}")
print(f"Vector v: {v}")
print(f"Vector u + v: {w}")
print(f"\n||u|| = {norm_u:.4f}, ||v|| = {norm_v:.4f}")
print(f"||u + v|| = {norm_w:.4f} <= ||u|| + ||v|| ({norm_u + norm_v:.4f})")
print(f"Triangle Inequality holds: {norm_w <= norm_u + norm_v}")
```

4. Key Takeaways & Interview Points

- **ResNet Residual Connections:** Deep learning models with hundreds of layers succeed because vector addition $\mathbf{x}_{\ell+1} = \mathbf{x}_{\ell} + \mathcal{F}(\mathbf{x}_{\ell})$ allows gradients to flow backwards without vanishing.
- **Commutative & Associative:** Vector addition is abelian: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ and $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.